

Supplementary Information

Discovering Molecular Substructures from Mass Spectra with Neural Topic Models

This document accompanies the [main paper](#). It preserves the full model equations, evaluation protocols and reproducibility record underlying the shorter explanations there. S-prefixed section, equation, figure and table numbers refer to this supplement. References to main figures open the companion PDF; keep both files in the same directory.

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S1 Notation and spectral data

S1.1 Notation and scope

Table S1 defines the main indices and quantities before the model equations. A word index identifies a *vocabulary type*; a token is one count-weighted copy of that word, not necessarily a distinct physical peak. Throughout, sums over w or v cover the vocabulary unless a channel or evaluation subset is specified, and sums over k or j cover all topics. Subscript d denotes a spectrum, never a dataset or training repetition.

Table S1: Core notation. Auxiliary quantities are defined where introduced.

Symbol	Meaning and dimensions
$D; d = 1, \dots, D$	Number of spectra in the stated partition; spectrum index.
$V; w, v = 1, \dots, V$	Vocabulary size; spectral-word indices.
$K; k, j = 1, \dots, K$	Number of fitted topics; topic indices.
$E; \mathcal{V}_F, \mathcal{V}_L$	Embedding dimension; disjoint fragment and loss vocabularies whose union is the full vocabulary.
$c_d, x_d \in \mathbb{R}^V; N_d$	Word pseudo-count vector, normalized encoder input, and total in-vocabulary count.
$\rho_w, \alpha_k \in \mathbb{R}^E$	Fixed word vector and learned topic vector; tables have shapes $V \times E$ and $K \times E$.
$\beta \in \mathbb{R}^{K \times V}; \theta_d \in \mathbb{R}^K$	Topic-word probabilities and spectrum-level topic proportions; each row/vector sums to one.
$z_d, m_d, \ell_d \in \mathbb{R}^K$	Gaussian latent vector, corrected posterior mean, and log variance.
$\phi; I_K$	MLP parameters; $K \times K$ identity matrix (not peak intensity).
$s_d, h_{dw} \in \mathbb{R}^E$	Whole-spectrum mean embedding and contextualized word direction.
$\lambda_{\text{ctx}} \in \mathbb{R}$	One learned context-strength scalar, distinct from count c_{dw} .
$r_d, o_d \in \mathbb{R}^K$	Pooled topic evidence and its centered-log mean correction.

Vector exponentials and logarithms operate elementwise; diag forms a diagonal matrix, and $\odot$ denotes elementwise multiplication. The notation $\mathcal{N}$ denotes a Gaussian distribution. A probability vector lies on the simplex: its entries are nonnegative and sum to one. Hats on embedding vectors mean unit normalization; hats on inferred mixtures mean deterministic estimates, as defined below.

S1.2 Spectral representation and data preparation

Real reference spectra. MSⁿLib is an open library of reference-standard fragmentation spectra, including multi-stage measurements [9]. We use the positive-ion-mode benchmark subset distributed in Mascot Generic Format (MGF) with the MS2LDA 2.0 data and models [10], not the entire MSⁿLib collection. An MGF record supplies a precursor m/z , a list of fragment m/z values and intensities, and metadata linking the spectrum to its known molecule. Known structures permit chemically grouped splitting and evaluation; they are not ground-truth labels for every latent motif.

Peak filtering and intensity representation. The input contains 41,568 spectra. Intensities are divided by the largest intensity within each spectrum; peaks outside $m/z \in (0, 2000]$ or normalized intensity $[0.01, 1]$ are removed. At most the 1,000 most intense peaks are retained, and spectra with fewer than five retained peaks are discarded. This leaves 38,888 spectra. Normalizing each spectrum to its strongest peak makes the representation independent of overall signal scale; the intensity cutoff suppresses weak peaks, the peak cap bounds document size, and the minimum-peak rule excludes spectra with too few peaks for the analysis. Weaker peaks may still carry structural information; these filtering choices define the benchmark input.

From peaks to spectral words. Each retained physical peak generates a fragment word and, if precursor-minus-peak m/z exceeds 0.01, a neutral-loss word. Here “loss” follows the benchmark’s precursor-minus-fragment convention; it does not infer charge-resolved neutral compositions. Mass values are rounded to two decimal places, for example **frag@70.04** and **loss@162.05**. Normalized peak intensity I contributes $\text{round}(100I)$ copies of each applicable word. Let c_{dw} be the resulting pseudo-count of word w in spectrum d and $N_d = \sum_w c_{dw}$. The normalized input is

$$x_{dw} = \begin{cases} c_{dw}/N_d, & N_d > 0, \\ 0, & N_d = 0. \end{cases}$$

The encoder receives x_d ; the reconstruction term uses raw counts c_d . Counts are summed when multiple peaks yield the same word. They encode discretized relative intensity, not independent ion counts or a calibrated noise model. For example, a peak at $m/z = 60.00$ from a precursor at $m/z = 100.00$ with $I = 0.40$ contributes 40

copies each of **frag@60.0** and **loss@40.0**, if both words are in the vocabulary. Token strings retain the rounded numerical value, without padding trailing zeros. Main Figure 1 shows the transformation and which steps use training data only.

Structure-aware splitting and training-only representation. Molecules are grouped by connectivity InChIKey and, for cyclic compounds, non-chiral Bemis–Murcko scaffold [11]; each acyclic connectivity group stays intact. A deterministic allocation of 28,572 groups toward a 70/10/20 split yields 27,222 training, 3,889 validation and 7,777 reserved test spectra, with zero compound or split-group overlap. We retain words occurring in at least three training spectra, yielding a vocabulary of $V = 21,233$ words. Keeping related structures within one partition reduces evaluation leakage from repeated compounds or shared scaffolds. Restricting the vocabulary to recurring training words removes extremely rare features without consulting evaluation data. In particular, validation spectra and chemical labels do not train the vocabulary, embeddings or model weights. The results here use validation, not the reserved test partition.

Every vocabulary word has a fixed embedding vector ρ_w of dimension $E = 48$, learned by skip-gram with negative sampling (SGNS) on training spectra [6]. Words observed together supply positive training pairs; frequency-weighted negative pairs provide contrasts. This makes embedding geometry reflect recurring spectral associations without structural labels. Source and context embeddings are averaged and normalized after training. These coordinates are held fixed during topic-model fitting and inference. The embedding-training settings are given in Section S8.2.

Controlled synthetic spectra. The simulator plants $R = 18$ motifs and activates one to three in each spectrum. Motifs have characteristic fragment/loss anchors, some of which are shared; precursor-dependent complementary peaks, background features, noise and variable intensities make recovery nontrivial. Spectra contain 18–41 physical peaks before conversion to paired words. Each realization contains 800 training and 160 validation spectra, with vocabulary filtering and word embeddings derived from its training portion. Ground-truth mixtures are normalized motif-labeled signal counts, excluding background/noise; ground-truth word distributions are normalized labeled training counts in the retained vocabulary. They are evaluation targets, not parameters supplied to the fitted model. The simulator tests statistical recovery under controlled structure, not mechanistic prediction of real molecular fragmentation.

S2 Detailed model formulation

S2.1 Base model: the published ETM

ETM explains an observed spectrum by combining a shared set of topics. There are two quantities to learn: what words each topic favors, and how strongly each topic contributes to each spectrum. The decoder defines the first; a neural encoder estimates the second from the observed word counts.

We use the published variant with pre-fitted word embeddings (ETM-PWE), keeping the spectral SGNS embeddings fixed during topic-model training [4].

Topic distributions. ETM represents topic k with a learned vector $\alpha_k \in \mathbb{R}^E$. Its standard emission probabilities are

$$\beta_{kw}^{\text{ETM}} = \frac{\exp(\alpha_k^\top \rho_w)}{\sum_{v=1}^V \exp(\alpha_k^\top \rho_v)}. \quad (\text{S1})$$

Generating a spectrum. Conditional on its length, a spectrum is generated in three steps: draw Gaussian topic scores, convert them to mixture weights, and draw words from the weighted topic distributions. Marginalizing the per-word topic assignment gives

$$z_d \sim \mathcal{N}(0, I_K), \quad \theta_d = \text{softmax}(z_d), \quad p(w \mid z_d) = \sum_{k=1}^K \theta_{dk} \beta_{kw}^{\text{ETM}}. \quad (\text{S2})$$

Conditional on length, the spectral document is multinomial with these word probabilities. Topics therefore retain a direct interpretation as distributions over spectral words. The inner product $\alpha_k^\top \rho_w$ is the unnormalized compatibility of topic k with word w . A change to a topic vector affects multiple word probabilities through the shared embeddings. In contrast, a model without this factorization estimates a full topic–word table. The embedding constraint shares information across related words, but also limits the emission patterns the model can represent.

Inferring the mixture from an observed spectrum. For observed counts, the topic scores are unknown. The encoder approximates their posterior with a diagonal Gaussian,

$$q_\phi(z_d | x_d) = \mathcal{N}(\mu_\phi(x_d), \text{diag}(\exp \ell_\phi(x_d))). \quad (\text{S3})$$

A two-hidden-layer MLP with rectified linear unit (ReLU) activations supplies a shared representation, followed by separate mean and log-variance heads. We use width 800. Reparameterization, $z_d = \mu_\phi(x_d) + \exp(\ell_\phi(x_d)/2) \odot \epsilon_d$ with $\epsilon_d \sim \mathcal{N}(0, I_K)$, enables gradient optimization [4].

Here $\mu_\phi(x_d)$ is the posterior mean, $\ell_\phi(x_d)$ is its log variance, and ϵ_d is independent standard-Gaussian sampling noise. The covariance specifies uncertainty in Gaussian latent space, not a calibrated probability of a chemical annotation. *Amortized inference* means that the same learned network predicts these quantities for every spectrum, including a new spectrum, without learning a separate set of parameters for that spectrum.

Fitting the model. Training fits the topic vectors and encoder together by maximizing the base-model evidence lower bound (ELBO):

$$\mathcal{L}_d^{\text{ETM}} = \mathbb{E}_{q_\phi(z_d | x_d)} \left[\sum_w c_{dw} \log \left(\sum_k \text{softmax}(z_d)_k \beta_{kw}^{\text{ETM}} \right) \right] - \text{KL}(q_\phi(z_d | x_d) \| \mathcal{N}(0, I_K)). \quad (\text{S4})$$

The first term rewards mixtures that reconstruct the observed words; the second regularizes the inferred Gaussian against its standard-normal prior. The data-constant multinomial coefficient is omitted. The encoder is an inference tool: the generative model remains a Gaussian draw, a topic mixture, and words sampled from that mixture.

For a concrete example, suppose a spectrum has topic weights (0.7, 0.3) and a particular fragment has probability 0.4 under topic 1 and 0.1 under topic 2. Its predicted probability is $0.7(0.4) + 0.3(0.1) = 0.31$. Our enhanced model keeps this weighted-sum interpretation. It changes the topic distributions, the inferred Gaussian mean, and the map from Gaussian scores to mixture weights, as explained next.

S2.2 Enhancement 1: balance fragment and loss probabilities

The first change is to the decoder. Fragments and losses are two complementary views of an observed peak, but the base ETM puts all their words into one softmax. A topic can therefore assign, for example, 85% of its probability to fragments and only 15% to losses. We give each channel 50% of every topic’s probability, while still learning which words matter *within* each channel. Main Figure 2a shows this example; panels (b) and (c) preview the next two changes.

For the fragment and neutral-loss vocabularies $\mathcal{V}_F$ and $\mathcal{V}_L$, replace the single normalizer of Equation S1 with

$$\beta_{kw} = \begin{cases} \frac{1}{2} \frac{\exp(\alpha_k^\top \rho_w)}{\sum_{v \in \mathcal{V}_F} \exp(\alpha_k^\top \rho_v)}, & w \in \mathcal{V}_F, \\ \frac{1}{2} \frac{\exp(\alpha_k^\top \rho_w)}{\sum_{v \in \mathcal{V}_L} \exp(\alpha_k^\top \rho_v)}, & w \in \mathcal{V}_L. \end{cases} \quad (\text{S5})$$

Each topic remains a probability distribution, with half its mass allocated to each channel. No additional learned parameter is introduced.

Fragments and losses provide complementary descriptions of a motif. Separate normalization prevents one channel from taking most of its total probability mass simply through competition across the combined vocabulary. Equivalently, generate the channel with probability 1/2, then generate a word using that channel’s ETM distribution. Since a word reveals its channel, the factor 1/2 contributes a constant $-\log 2$ per token to the conditional log-likelihood; learning concerns the within-channel emissions. Equal channel mass is an explicit assumption, not an estimate of the observed channel ratio, and does not force equal token counts or gradient contributions.

S2.3 Enhancement 2: use the surrounding spectrum to guide inference

The second change is to the encoder’s Gaussian mean. A fragment can support different topics depending on which other fragments and losses accompany it. The base ETM’s MLP already sees the whole count vector. We add an explicit embedding-based route: combine each word with a summary of its spectrum, match that contextual direction to the decoder’s topic vectors, and pool the matches into a correction to the MLP mean. The MLP still predicts the uncorrected mean and log variance; the added route supplies topic evidence using the same embedding geometry as the decoder.

Main Figure 2b gives a two-dimensional example. The same fragment points along (1, 0) in spectra A and B. With context strength one, mean embeddings (0.5, 0.5) and (0.5, −0.5) tilt its matching direction above or below

that shared word vector. Thus the word can favor a different topic in each spectrum even though its stored embedding is unchanged. These are illustrative coordinates, not measured spectra or chemical assignments.

Formally, let $\hat{\rho}_w = \text{normalize}(\rho_w)$ and $\hat{\alpha}_k = \text{normalize}(\alpha_k)$, where normalization divides by the Euclidean norm floored at 10^{-12} . First compute one mean embedding for the full spectrum and combine it with each observed word:

$$s_d = \sum_w x_{dw} \hat{\rho}_w, \quad h_{dw} = \text{normalize}(\hat{\rho}_w + \lambda_{\text{ctx}} s_d). \quad (\text{S6})$$

The same s_d is used for every word, including the word being scored. The context strength λ_{ctx} is a single learned, unconstrained scalar initialized to one; there is no separate context network.

For $N_d > 0$, let T_{dw} contain the two topics with largest cosine scores $a_{dwk} = h_{dw}^\top \hat{\alpha}_k$. A unit-temperature local softmax gives evidence weights, which are then pooled:

$$\pi_{dwk} = \begin{cases} \frac{\exp(a_{dwk})}{\sum_{j \in T_{dw}} \exp(a_{dwj})}, & k \in T_{dw}, \\ 0, & k \notin T_{dw}, \end{cases} \quad r_{dk} = \sum_w x_{dw} \pi_{dwk}. \quad (\text{S7})$$

The pooled vector has nonnegative entries summing to one; numerical renormalization removes accumulation error. An empty-input row receives uniform evidence. Here π_{dwk} denotes attention weights, not a second latent variable or an exact word-topic posterior.

Convert the pooled evidence into a centered log offset:

$$o_{dk} = \log(1 + K r_{dk}) - \frac{1}{K} \sum_{j=1}^K \log(1 + K r_{dj}), \quad m_d = \mu_\phi(x_d) + o_d. \quad (\text{S8})$$

This equals the centered form of $\log(r_{dk} + 1/K)$ because the constant $\log K$ cancels. Uniform evidence gives zero offset; the fixed pseudocount $1/K$ keeps every logarithm finite. Centering removes a common shift that cannot affect the subsequent simplex mixture. The final posterior is $\mathcal{N}(m_d, \text{diag}(\exp \ell_\phi(x_d)))$.

Two examples illustrate the mean correction. Uniform evidence $r_{dk} = 1/K$ leaves the MLP mean unchanged. For three topics with $r_d = (1, 0, 0)$, the offset is $(\frac{2}{3} \log 4, -\frac{1}{3} \log 4, -\frac{1}{3} \log 4)$: the supported topic receives a positive relative adjustment, and unsupported topics remain finite. The MLP retains flexibility to learn corrections and uncertainty beyond the matching rule. Top-2 matching avoids a hard top-1 decision while keeping local evidence selective; it does not restrict the entire spectrum to two topics. The choice of two is a modeling assumption, and interpreting these attention weights chemically requires further evidence. The hard top-2 choice can switch discontinuously at rank ties; gradients follow the selected branch away from such ties. Training remains nonconvex, and the experiments do not establish topic identifiability.

S2.4 Enhancement 3: let unsupported topic weights become zero

The third change is the map from Gaussian scores to topic weights. Softmax gives every topic a positive weight, including topics with very weak scores. We use 1.5-entmax so that a spectrum can be explained by a subset of topics, with exactly zero weight on the others.

For the same illustrative latent scores $z = (1, 0, -1)$, softmax yields approximately $(0.665, 0.245, 0.090)$, whereas 1.5-entmax yields $(0.831, 0.169, 0)$ (main Figure 2c). The weakest component is excluded without applying a post-hoc weight cutoff. This example explains the mapping; it does not predict how many topics a fitted spectrum will use.

The published 1.5-entmax map replaces softmax [8]:

$$\theta_{dk} = \text{entmax}_{1.5}(z_d)_k = \left[\frac{z_{dk}}{2} - \tau(z_d) \right]_+^2, \quad \sum_k \theta_{dk} = 1, \quad (\text{S9})$$

where $[u]_+ = \max(u, 0)$ and τ is fixed by normalization. This map can assign exact zeros to weakly supported topics and is differentiable almost everywhere. It imposes neither an auxiliary sparsity loss nor a fixed spectrum-level support size.

The intended benefit is a compact explanation of each spectrum without forcing all spectra to use the same small subset of topics. Unlike softmax, entmax changes the induced prior on the simplex: z_d still has a Gaussian prior, but θ_d is no longer logistic-normal and may lie on simplex faces. Replacing softmax with entmax therefore changes ETM's generative model, rather than only its inference procedure.

S2.5 Putting the three enhancements together

Starting from the base ETM in Section S2.1, we make exactly three substitutions: use the balanced β from enhancement 1 in place of β^{ETM} ; add enhancement 2's offset o_d to the posterior mean; and replace softmax with

enhancement 3’s 1.5-entmax. The fixed word embeddings, learned topic vectors, two-layer MLP, Gaussian prior and reconstruction-minus-KL form of the objective remain in place. Enhancements 1 and 3 change the generator; enhancement 2 changes the variational inference pathway.

The complete generator draws $z_d \sim \mathcal{N}(0, I_K)$, sets $\theta_d = \text{entmax}_{1.5}(z_d)$ and emits words with $p(w | z_d) = \sum_k \theta_{dk} \beta_{kw}$ using Equation S5. Its evidence lower bound (ELBO), omitting only the data-constant multinomial coefficient, is

$$\mathcal{L}_d = \mathbb{E}_{q(z_d|x_d)} \left[\sum_w c_{dw} \log \left(\sum_k \text{entmax}_{1.5}(z_d)_k \beta_{kw} \right) \right] - \text{KL}(q(z_d | x_d) \| \mathcal{N}(0, I_K)), \quad (\text{S10})$$

$$\text{KL} = \frac{1}{2} \sum_k [m_{dk}^2 + \exp(\ell_{dk}) - 1 - \ell_{dk}]. \quad (\text{S11})$$

Here $\mathcal{L}_d$ is the per-spectrum lower bound, $\ell_d = \ell_\phi(x_d)$, and $q(z_d | x_d)$ is the corrected Gaussian posterior from Equation S8. Maximizing the sum of these bounds over training spectra balances reconstruction of their observed words against departure of the posterior from the standard-Gaussian prior. The pseudo-count scale fixes the relative strength of reconstruction and KL regularization: changing it changes the fitted objective even when normalized encoder inputs are unchanged. The KL uses the corrected mean m_d , not the unmodified MLP mean. Sharing topic vectors between inference and decoding does not invalidate this bound. It is defined in Gaussian latent space and therefore needs no invertible entmax map or entmax Jacobian term.

Equation S10 states the mathematical objective. In the implementation, the reconstruction term uses $\log \max(p, \varepsilon)$ with $\varepsilon = 10^{-12}$ to guard against numerical underflow. This is a numerical approximation: the floor does not define a normalized likelihood, and the resulting expression is not guaranteed to retain the exact lower-bound interpretation. Adam also applies coupled weight decay 1.2×10^{-6} to learned parameters; this optimizer regularization is distinct from the latent Gaussian KL and introduces no chemical-quality loss.

Main Figure 3 shows posterior construction followed by a worked topic-mixture calculation. The MLP supplies the posterior mean and variance, while context contributes an additive mean correction. Topic vectors enter the evidence pathway as unit directions and the decoder as raw inner products, linking both pathways without a second topic dictionary.

Real fits use $K = 1,000$, two hidden MLP layers of width 800, 120 epochs, batch size 200 and Adam optimization. The batch-mean negative ELBO uses raw-count reconstruction, KL weight one and one reparameterized sample per spectrum per step. There is no KL annealing, auxiliary motif-quality loss, batch normalization or spectrum-specific refinement. At $V = 21,233$, there are 19,278,001 trainable parameters, excluding the fixed embedding table. The context pathway adds only the scalar λ_{ctx} to the ETM parameterization. The full optimizer and embedding-training settings are given in Section S8.2.

Inference uses $\hat{\theta}_d = \text{entmax}_{1.5}(m_d)$ without fitting parameters to each new spectrum. It evaluates entmax at the posterior mean rather than averaging over Gaussian samples, so it is neither $\mathbb{E}_q[\text{entmax}_{1.5}(z_d)]$ nor an integrated posterior predictive distribution. Exported probabilities are renormalized to control floating-point error. Chemical evaluation uses full spectra. Completion supplies only observed peak groups to the entire inference pathway, including its context mean.

S3 Comparator and training protocols

S3.1 Correspondence with Tomotopy LDA

Tomotopy is a software library; LDA is the probabilistic model fitted by its LDA backend [13, 3]. In LDA, each topic row β_k has a Dirichlet prior with positive concentration vector $\eta \in \mathbb{R}^V$, and each spectrum mixture θ_d has a Dirichlet prior with concentration vector $\gamma \in \mathbb{R}^K$. For token position $n = 1, \dots, N_d$, a discrete topic assignment t_{dn} is drawn from θ_d , then the observed word w_{dn} is drawn from $\beta_{t_{dn}}$. Marginalizing t_{dn} gives the same additive word probability $\sum_k \theta_{dk} \beta_{kw}$ used by ETM. The symbols γ and α_k have different roles: the former controls LDA proportions, whereas the latter is an ETM topic embedding.

Tomotopy estimates assignments using collapsed Gibbs sampling. A token’s conditional topic probability depends on topic counts in its spectrum and word counts within each topic, together with the prior concentrations. Consequently, LDA already uses the rest of a spectrum when interpreting a word, and Dirichlet priors can favor concentrated mixtures. Co-occurrence, neutral losses and concentrated mixtures are therefore already part of the LDA-based approach.

The differences in Table S2 concern *how* these properties are represented and inferred. ETM introduces embedding-constrained emissions and a neural Gaussian posterior. Our enhancements further constrain channel mass, add a direct embedding-conditioned posterior pathway, and change the Gaussian-to-simplex mapping. Tomotopy uses iterative inference for held-out spectra; the neural model uses the fitted recognition function. This is an architectural distinction, not a measured speed advantage in this benchmark.

Table S2: Corresponding components in Tomotopy LDA, published ETM and the selected spectral enhancement. All retain additive topic-mixture emissions.

Component	Tomotopy LDA	Published ETM	Contextual Sparse ETM
Topic words	Probability table with Dirichlet prior	Embedding inner products and word softmax	Same embedding decoder; separate fragment/loss softmaxes with mass 1/2 each
Topic proportions	Dirichlet-distributed spectrum mixtures	Standard Gaussian latent vector mapped through softmax	Standard Gaussian latent vector mapped through 1.5-entmax
Inference	Gibbs updates informed by spectrum–topic and topic–word counts	MLP predicts Gaussian mean and log variance	Retain MLP; add whole-spectrum evidence to the mean
Parameter fitting	Sampling-based estimation of topic assignments and distributions	Gradient optimization of a variational lower bound	Same latent-space bound form with the stated modifications
New spectrum	Iterative topic inference with fitted topics	Reuse the learned encoder	Reuse encoder and contextual evidence pathway

S3.2 Experimental design and evaluation

Primary comparison: Tomotopy. The selected model and Tomotopy use the same training spectra, vocabulary, $K = 1,000$, validation split and chemical scoring procedure. The Tomotopy 0.13.0 comparator [13] uses initial $\gamma_k = 0.6$ and fixed $\eta_w = 0.1$ for every topic and word. The mixture concentrations are optimized every ten iterations. Training stops when absolute relative perplexity change is below 0.005 for ten consecutive 50-iteration checks, with at most 5,000 iterations. All three repetitions met this rule at 750–800 iterations. Held-out inference uses 100 iterations. Both models have three independently initialized fits on the same data. The three Tomotopy runs use the same training settings; the earlier single-run reference remains in the historical evidence and is not counted as an additional repetition. We hold the data fixed but do not match computational budgets or compare hyperparameter searches under an equal budget.

Secondary check: the ETM base. We also compare the enhanced model with its published ETM base. It shares the training embeddings, two-hidden-layer width-800 MLP, optimizer settings and 120-epoch budget, but uses batch size 256 rather than 200. Its three fits comprise the existing reference and two additional initializations with the same training settings. Keeping the original batch size allows us to summarize all three fits together, although it leaves a training difference from the selected model. This secondary comparison assesses the enhancements together rather than isolating each one’s effect.

Repetition and reporting. For each synthetic realization, the selected model is fitted with $K = 36$ and $K = 128$, both exceeding the 18 planted motifs. Three realizations vary the generated data and training initialization. In contrast, the three MSⁿLib fits of each model vary training initialization on one fixed data partition, with the neural models sharing fixed embeddings. Results are means $\pm$ sample standard deviations (SD) over the relevant three fits. The original plain-ETM fit used WSL2; its new repetitions use native Linux with matching scientific package versions (Section S8.2). Real-data SDs describe variation between training runs on the same data; synthetic SDs also include variation in the generated data and embeddings. Neither represents a confidence interval for differences between models or uncertainty across new chemical populations. Exact random states are recorded only in Section S8.

S4 Evaluation metric definitions

The metrics answer different questions and are not combined into one score. Synthetic recovery measures agreement with planted signal; completion assesses predictive fit; mixture summaries characterize topic use; MAG/SOS assesses reference-supported chemical coherence. The following definitions apply to all result tables.

Synthetic word and mixture recovery. Let $u = 1, \dots, R$ index the $R = 18$ planted motifs, and let β_u^* and θ_d^* denote the ground-truth word distribution and full-spectrum mixture defined by the simulator. Word truth is restricted to the retained training vocabulary; mixture truth uses all motif-labeled physical peaks, including out-of-vocabulary signal, and excludes background and noise peaks. Topic labels are arbitrary, so we first find the one-to-one assignment $\sigma(u)$ from planted motifs to fitted topics that maximizes the sum of

word-distribution cosine similarities, using the Hungarian algorithm. For nonzero vectors a, b , cosine similarity is $\cos(a, b) = a^\top b / (\|a\|_2 \|b\|_2)$. The reported measures are:

- *Word recovery*: $R^{-1} \sum_{u=1}^R \cos(\beta_u^*, \beta_{\sigma(u)})$. It evaluates learned motif emissions.
- *Mixture recovery*: mean across full validation spectra of $\cos(\theta_d^*, \tilde{\theta}_d)$, where $\tilde{\theta}_{du} = \hat{\theta}_{d, \sigma(u)}$ is the inferred mixture restricted to matched topics. Numerically zero aligned mass receives score zero.
- *Recovered motifs*: number of planted motifs whose matched word cosine is at least 0.50, reported out of 18.

Both cosine averages lie in $[0, 1]$ here and larger values indicate better agreement. The recovery-count threshold is a reporting convention, not an identifiability guarantee. Mixture recovery omits unmatched fitted topics; it is not a penalty on all extraneous topic mass. The continuous word score and the count therefore need to be read alongside the mixture score.

Document-completion negative log-likelihood. Document completion divides each validation spectrum 50/50 by whole physical peak groups. A fragment and its corresponding loss stay together. For real spectra, the observed peaks are normalized and discretized without reference to withheld intensities. The synthetic benchmark instead partitions already discretized full-spectrum peak counts, then normalizes the observed count row for the encoder. Its observed integer counts therefore retain a rounding dependence on the original full-spectrum intensity maximum; the synthetic completion construction is not identical to the real-data construction. Only in-vocabulary withheld pseudo-tokens are scored. With the numerical floor $\varepsilon = 10^{-12}$, the reported metric is

$$\text{NLL}_{\text{comp}} = - \frac{\sum_{d \in \mathcal{D}_{\text{comp}}} \sum_{w=1}^V c_{dw}^{\text{held}} \log \max \left(\sum_k \hat{\theta}_{dk}^{\text{obs}} \beta_{kw}, \varepsilon \right)}{\sum_{d \in \mathcal{D}_{\text{comp}}} \sum_{w=1}^V c_{dw}^{\text{held}}}. \quad (\text{S12})$$

Here $\mathcal{D}_{\text{comp}}$ is the set of eligible completion spectra, c_{dw}^{held} is the withheld count, and $\hat{\theta}_d^{\text{obs}}$ is inferred using only the observed half. For neural models, this is the deterministic plug-in estimate defined in Section S2, not an integral over posterior uncertainty. The denominator is the number of scored token copies, so this is a token-weighted average, not an unweighted average of spectrum-level scores. Lower values indicate better prediction, in nats per withheld pseudo-token (natural logarithms). The selected-model denominator is 1,277,983 in-vocabulary tokens from 3,888 eligible validation spectra. Eligibility requires positive withheld in-vocabulary mass, not positive observed mass. One eligible real spectrum has no observed in-vocabulary words: neural inference uses its zero-input MLP posterior (and uniform contextual evidence, hence zero context offset), whereas Tomotopy uses its learned, normalized Dirichlet prior. A spectrum with no withheld in-vocabulary mass is excluded. A further 3.14% of withheld token copies are out of vocabulary and excluded. The corresponding exclusion is 35.42–37.33% across the three synthetic realizations, so their completion scores describe only the retained vocabulary. Vocabulary and tokenization affect this metric: values can be compared between models on the same benchmark, not directly between the synthetic and real datasets.

Local sparsity and corpus-wide topic use. The entropy-effective topic count is

$$K_{\text{eff}}(\theta_d) = \exp \left[- \sum_k \theta_{dk} \log \theta_{dk} \right], \quad (\text{S13})$$

with $0 \log 0 = 0$. The implementation evaluates positive entries using $\log \max(\theta_{dk}, 10^{-12})$ for numerical stability. The ideal count ranges from one for a single-topic mixture to K for a uniform mixture. We compute the median over full validation spectra within each fit before averaging those medians across repetitions. *Exact support* instead counts indices k with $\hat{\theta}_{dk} > 0$; tiny positive weights count even if their contribution to entropy is negligible. *Dominant-topic count* is the number of distinct topics attaining the largest weight in at least one validation spectrum, with ties resolved by topic index. These separate local compactness from global topic collapse. A smaller effective count is not intrinsically better if it results from discarding genuine structure or using very few topics throughout the corpus.

MAG: linking spectral motifs to structural hypotheses. Mass2Motif Annotation Guidance (MAG) is the MS2LDA 2.0 procedure for suggesting structural interpretations of a learned motif [2]. It converts the topic’s highest-probability words into a motif *pseudo-spectrum*, retrieves similar reference spectra, and looks for consistent evidence among their known molecules. Thus MAG is downstream annotation, not a neural training objective or a second topic model. The annotation library excludes every row sharing a validation or test connectivity InChIKey. Each topic’s 20 highest-probability words form a motif pseudo-spectrum. Spec2Vec retrieval searches up to 1,000 neighbors and retains at most ten unique molecules. For refinement, MAG compares each retrieved spectrum with masked versions of the motif and clusters the resulting similarity-profile vectors. The inherited implementation applies complete-linkage Euclidean clustering to one minus these vectors,

with distance threshold $1 - 0.9 = 0.1$, and retains the cluster containing the highest-ranked retrieval hit. The parameter named cosine similarity in the software is therefore not a direct pairwise spectral-cosine cutoff. A motif is *optimized* when at least one fragment or loss survives refinement. Binary MACCS (Molecular ACCess System) fingerprints encode predefined molecular features; bits present in at least 80% of the annotation cluster define consensus feature set A_k [12].

SOS: agreement with the motif-associated molecules. The substructure overlap score (SOS) asks how much of the MAG consensus fingerprint occurs in the known molecules associated with a motif. Retrieval candidates used to form the consensus and validation molecules used to evaluate it have different roles; the library exclusion above prevents an evaluated compound from supplying its own retrieval evidence.

Instead of the original MS2LDA 2.0 posterior-weight threshold of 0.5, each validation spectrum is associated with its largest full-spectrum topic weight, resolving ties by topic index. This gives exactly 3,889 associations per model without assuming comparable calibration of absolute topic weights. Repeated connectivity InChIKeys are collapsed within a topic. Let C_k be the set of unique associated molecules with valid fingerprints, let $i \in C_k$ index a molecule, and let B_i be its set of active fingerprint features. Vertical bars denote set cardinality. Then

$$\text{SOS}_k = \frac{1}{|C_k|} \sum_{i \in C_k} \frac{|A_k \cap B_i|}{|A_k|}. \quad (\text{S14})$$

Empty A_k receives overlap zero. For example, if A_k contains five features and an associated molecule contains four of them, its overlap contribution is $4/5 = 0.8$. SOS averages these contributions across C_k and lies in $[0, 1]$; larger values indicate greater fingerprint containment. It does not mean that 80% of a molecule’s atoms or bonds were identified correctly, and it does not require the molecule to contain only the consensus features. An optimized motif is *evaluable* when a MAG consensus fingerprint was successfully constructed and C_k is nonempty. A valid all-zero consensus receives SOS zero; a missing consensus is not evaluable. A motif is *useful* when additionally $\text{SOS} \geq 0.6$, the intermediate-or-high MS2LDA 2.0 band. Mean SOS averages only evaluable motifs, hence different models can be averaged over different motif populations. We report both the number of evaluable motifs and the number meeting the usefulness cutoff, together with mean SOS. The cutoff is an operational screening convention, not a sharp boundary between true and false chemistry. SOS measures fingerprint containment, not atom-level identification or a fragmentation-pathway proof.

S5 Chemical specificity and correspondence protocols

We supplement MAG/SOS with four post-hoc assessments using the same three selected-model and three Tomotopy fits. No model is retrained and no MAG annotation is changed. Plain ETM remains the enhancement baseline in the main comparison, but is omitted from these additional analyses because the full topic and mixture arrays for one of its three fits are unavailable locally. The new analyses use the development-validation data, not an independent confirmation set. We fixed the protocol and input hashes before calculating these additional outcomes; this does not undo earlier model selection on the same data.

S5.1 Compound support and matched-background SOS

The 3,889 validation spectra represent 3,854 connectivity compounds. We retain the dominant-topic association and within-topic compound deduplication in Equation S14. A compound with multiple spectra can support more than one topic, but its fingerprint remains the same in every occurrence. We report compound and scaffold support for all 1,000 topics in each fit; a single-compound motif is a candidate, not evidence of recurring chemistry.

The primary background has 158 strata, including 64 compounds in fixed singleton strata.

Raw SOS can be high when the annotation contains features common in many molecules. To estimate this background, we partition compounds into strata. Let s index a stratum, G_s its compound set, $N_s = |G_s|$, and $n_{ks} = |C_k \cap G_s|$. Strata combine each compound’s acquisition profile (the set of its adduct–collision-energy pairs) with $\lfloor \tilde{m}_i/50 \rfloor$, where $\tilde{m}_i$ is its median precursor mass-to-charge value in Da for singly charged ions. All spectra here are positive-ion Orbitrap HCD MS2 spectra. Raw acquisition fields are joined to the saved validation records by exact universal spectrum identifier (USI), with complete coverage. With $g_{ki} = |A_k \cap B_i|/|A_k|$ and the same empty-consensus convention as SOS, the conditional background and excess are

$$\mu_{0k} = \frac{1}{|C_k|} \sum_s n_{ks} \frac{1}{N_s} \sum_{i \in G_s} g_{ki}, \quad \Delta_k = \text{SOS}_k - \mu_{0k}. \quad (\text{S15})$$

Thus Δ_k measures fingerprint agreement beyond that expected for compounds with the matched attributes. The null population includes the motif’s own compounds, as required by the conditional randomization scheme. It does not ask whether precursor mass itself carries chemical information.

We draw $B = 99,999$ random permutations of complete compound fingerprints within strata, sharing each permutation across all six fits and preserving correlations among fingerprint bits. If b_k permuted scores are at least the observed score, the one-sided value is $p_k = (1 + b_k)/(1 + B)$ [18]. Integer intersection counts give exact inclusive tie handling. The evidence records exact binomial intervals for Monte Carlo tail-probability precision, separately from between-fit variation. Singleton strata cannot move; zero-variance nulls are labelled uninformative. Missing annotations are not assigned an observed SOS of zero.

Sensitivity analyses use 25 and 100 Da bins, and a separate 50 Da analysis with one compound per nonempty scaffold. The scaffold representative has the smallest SHA-256 hash of its connectivity key, independently of model scores; acyclic compounds remain distinct. This yields the same 2,856-compound subset for every fit. Strata are rebuilt within that subset.

S5.2 Enrichment of explicit molecular features

Independently of MAG availability, we test all 166 native RDKit MACCS positions for every supported motif. Position zero is unused; overlapping MACCS predicates are molecular features, not 166 unique substructures. The pinned definitions, SMARTS patterns and procedural exceptions are included in the evidence. Let $f \in \{1, \dots, 166\}$ index a key, $Y_{if} = \mathbb{1}\{f \in B_i\}$ its presence in compound i , $M_{sf} = \sum_{i \in G_s} Y_{if}$ the stratum-positive count, and $X_{kf} = \sum_{i \in C_k} Y_{if}$ the motif-positive count. Under the same conditional null, the contribution from stratum s is hypergeometric:

$$X_{ksf}^{(0)} \sim \text{Hypergeom}(N_s, M_{sf}, n_{ks}), \quad \delta_{kf} = \frac{X_{kf} - \sum_s n_{ks} M_{sf} / N_s}{|C_k|}. \quad (\text{S16})$$

We convolve the finite stratum distributions to obtain the exact one-sided count tail, without a large-sample approximation. δ_{kf} is the prevalence difference from matched background. The corresponding prevalence ratio is recorded only when the background is nonzero. A feature association does not establish which atoms generated a spectral fragment.

For each fixed background configuration, Benjamini–Yekutieli adjustment [19] is applied separately to all 6,000 SOS slots and all 996,000 motif–feature slots across the six fits. Missing, empty and constant tests contribute $p = 1$ to the declared family. We show effect distributions and use $q \leq 0.05$ only as a descriptive screen. These are conditional, exploratory values: dependence-robust multiplicity correction does not repair model-selection bias or guarantee exchangeability within acquisition/mass strata. Sensitivity configurations are reported separately, not searched for the most favorable significance result.

S5.3 Repeatability, redundancy and annotated reference correspondence

We match all topics using cosine similarity of the full, common-vocabulary β rows and a Hungarian one-to-one assignment. Comparisons comprise three within-model fit pairs for each model and all nine cross-model pairs. We retain every matched score, including weak forced matches. Complementary measures are reciprocal-nearest-neighbor agreement, top-20-word Jaccard, supporting-compound Jaccard, and cosine of the signed feature-effect vectors δ_k . Jaccard is intersection size divided by union size; an empty union or a zero effect-vector norm is undefined. Nearest-neighbor ties use the lowest topic index. Within each fit, the closest other β row and exactly repeated available MAG consensus fingerprints describe redundancy. These pairwise comparisons share fits and are not independent replicates.

For reference correspondence, we use the locally available positive-ion GNPS, MassBank and Urine MotifDB sets [14]. Identical typed mass–intensity signatures, after joint maximum-intensity normalization, are deduplicated while retaining every source identity and conflicting manual label. The export’s paired NaN channel padding is treated as missing data; malformed spectra and zero/nonfinite embeddings are explicitly unscorable. Automated annotation strings are not used as reference truth. The existing Spec2Vec model embeds each reference and the learned topic’s 20 highest-probability words, with two-decimal fragment/loss words and square-root intensity weights. We report the ranked cosine scores and vocabulary coverage without defining a chemical-correctness cutoff. Direct overlap counts use maximum-cardinality one-to-one mass matching, separately for fragments and neutral losses, at 0.01 Da tolerance, with 0.005 and 0.02 Da sensitivities. Both query and reference feature counts are retained as denominators. MotifDB correspondence is corroborative, partly correlated with MAG, and neither an expert confirmation nor a test of chemical novelty.

S6 Directed recovery between model families

To measure how much of each inventory has a counterpart in another fit, we supplement one-to-one matching with a directed nearest-neighbor analysis. Let a and b index distinct fitted runs, $\mathcal{I}_a$ and $\mathcal{I}_b$ their selected topic-index sets, and $\beta_k^{(a)}$ the full word-probability vector for topic k in fit a . All vectors use the same 21,233

ordered vocabulary coordinates. We define

$$s_{k\ell}^{ab} = \frac{\beta_k^{(a)} \cdot \beta_\ell^{(b)}}{\|\beta_k^{(a)}\|_2 \|\beta_\ell^{(b)}\|_2}, \quad m_k^{a \rightarrow b} = \max_{\ell \in \mathcal{I}_b} s_{k\ell}^{ab},$$

$$C_{a \rightarrow b}(t) = \frac{1}{|\mathcal{I}_a|} \sum_{k \in \mathcal{I}_a} \mathbf{1}\{m_k^{a \rightarrow b} \geq t\}, \quad 0 \leq t \leq 1. \quad (\text{S17})$$

Here $\mathbf{1}\{\cdot\}$ is one when its condition holds and zero otherwise; t is a similarity threshold and $C_{a \rightarrow b}(t)$ is the fraction of source motifs with at least one target counterpart meeting it. Target motifs may be reused; exact nearest-neighbor ties choose the lowest topic index. This definition keeps every source motif in the denominator, including weak matches, and does not force a unique counterpart. Cosine itself is a similarity, not a percentage recovered or a probability of chemical identity.

We evaluate all 1,000 topics and, separately, recurring, evaluable motifs: those with at least two distinct supporting compounds and an available MAG consensus. Both source and target inventories are filtered *before* searching. A sensitivity cohort requires recurrence without MAG availability. Selecting only qualifying pairs after matching would answer a different question and could omit difficult-to-match source motifs. A second sensitivity uses the arithmetic mean of fragment-only and loss-only cosines, each independently normalized, to give both spectral channels equal influence. All saved topics have nonzero probability mass in both channels.

The second-largest score, $m_k^{(2),a \rightarrow b}$, determines the fraction $M_{a \rightarrow b}(t) = |\mathcal{I}_a|^{-1} \sum_{k \in \mathcal{I}_a} \mathbf{1}\{m_k^{(2),a \rightarrow b} \geq t\}$ with at least two counterparts. The fractions with zero, exactly one, or multiple counterparts are therefore $1 - C$, $C - M$, and M . We show these relationships across 201 equally spaced thresholds from zero to one, not at a selected chemical-correctness cutoff. Multiple spectral neighbors can suggest alternative descriptions of a pattern, but do not by themselves establish a chemical split or merge.

For the closest spectral counterpart, we also record separate fragment/loss cosines, top-20-word Jaccard, supporting-compound Jaccard, and cosine of the 166-dimensional signed feature-effect vector $\delta_k = (\delta_{kf})_{f=1}^{166}$ from Eq. S16. An empty set union or zero effect-vector norm is undefined, not zero agreement. Chemical agreement never enters the selection of the spectral counterpart.

All ordered fit pairs are evaluated: six within each model and nine in each cross-model direction. We calculate each fit-pair statistic separately, average target fits within each source fit, and report the mean and sample SD over three source-fit summaries. For a score, the fit-pair statistic is its median across source motifs; recovery curves average the corresponding fractions. We never pool target runs into one larger search library or choose the most favorable target run. Shared fits make these summaries dependent; the SD describes variation, not a confidence interval or an equivalence test. Examples are selected by a fixed score-quantile rule among recurring, evaluable cross-model queries: nearest to the pooled median and 90th percentile of best scores, plus the neural query nearest the 90th percentile of second-best scores. Their typed top-20 words and direct mass overlaps are retained for inspection. These are illustrative examples, not a sample of chemically confirmed motifs.

S7 Additional correspondence diagnostics

These diagnostics support the main paper’s comparison of spectral and chemical agreement (main Section 4.3). Figure S1 shows the feature-agreement distributions, multiple-counterpart fractions and channel-weight sensitivity for the same recurring, evaluable cohorts. The method and aggregation are defined in Section S6.

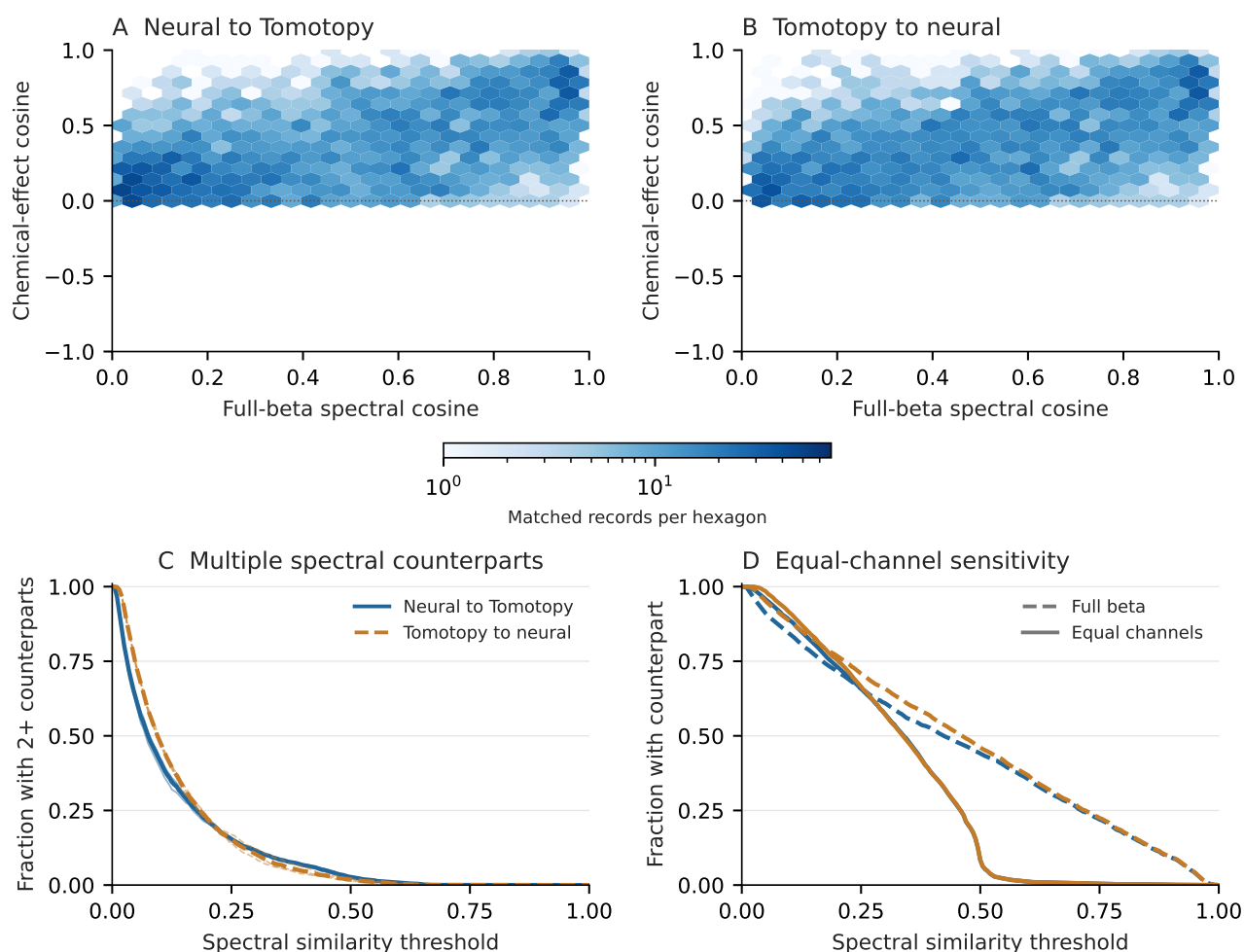


Figure S1: What cross-model correspondence does and does not establish. All panels use recurring, evaluable source and target inventories. (A–B) Full- β cosine versus signed MACCS feature-effect cosine for the closest counterpart; hexagon shading counts fit-pair/topic records on a shared logarithmic scale. Reused targets and repeated fits make these records dependent. A includes 4,631 records and excludes 1 undefined profile; B includes 4,521 and excludes 3. These exclusions apply only to chemical agreement, not spectral recovery. (C) Fraction of source motifs with at least two counterparts above each threshold; thin lines show source-fit variation. (D) Mean recovery with full- β cosine (dashed) and equal-weight fragment/loss cosine (solid), with the same directional colors as C. Spectral multiplicity is not proof of chemical splitting or merging, and the similarity scales are not probabilities of correct chemistry.

S7.1 Redundancy within each fitted inventory

Within-fit redundancy also tempers inventory counts. The mean median cosine to the nearest other topic is 0.093 for the selected model and 0.025 for Tomotopy. Exact MAG consensus profiles recur for 96.3 and 47.7 topics per fit, respectively (11.9% and 7.8% of available annotations). These are repeated coarse fingerprints, not proof that the corresponding motifs are chemically identical. They nevertheless show why motif count cannot be read directly as a count of distinct substructures.

S8 Reproducibility and scope of component evidence

S8.1 Implementation and evidence

Research code, manuscript sources and versioned evidence are available in the [public research branch of the MS2LDA fork](#). The selected implementation is `reduced_document_context` in `benchmarks/neural_ms2lda/contextual_reductions.py`, with shared mathematical functions in `contextual_sparse_etm.py` and training through `scripts/run_minimal_etm.py`. It remains separate from production Tomotopy and the archived leave-one-out model.

Historical single-fit reference evidence remains under `research/contextual_sparse_etm_msnlib/evidence/20260901_clean_room/`. The current three-fit baseline comparison, input identities and source manifest are under `research/baseline_repeats_20260908/evidence/`.

Selected-model evidence consists of nine completed fits, identified by the selected implementation name in `research/minimal_neural_etm/review_20260907/evidence/within_model/runs.csv`. Use recorded source snapshots for exact historical replay. A later gradient correction affects only uniform evidence, a case proved unreachable for the real and synthetic $K = 128$ training inputs but not the $K = 36$ trajectories. Saved results are unchanged; the qualification is documented in `docs/research/equation_code_audit_20260908.md`.

The additional assessment protocol, input seal, full evidence and reproduction commands are in `research/motif_chemical_assessment_20260908/`. Equations [S15](#) and [S16](#) are implemented in `benchmarks/neural_ms2lda/chemical_nulls.py`, with exhaustive small-population checks. Directed recovery (Equation [S17](#)) is implemented in `benchmarks/neural_ms2lda/cross_model_overlap.py`; its sealed evidence, full replay and denominator audit are in `research/cross_model_overlap_20260908/`. Neither assessment changes a saved fit or production model.

S8.2 Training settings and random states

The canonical environment is `ms2lda-neural`, specified by the root `environment.yml`; training used PyTorch 2.10.0 with CUDA 12.8. SGNS uses five epochs, eight count-weighted positive pairs per training spectrum and five negatives per pair. Negatives follow corpus frequency raised to 0.75. SparseAdam uses learning rate 0.01 and batches of 8,192 pairs. Neural topic-model training uses Adam learning rate 0.005, moments (0.9, 0.999) and weight decay 1.2×10^{-6} , with 120 epochs and batch size 200. The plain-ETM reference uses batch size 256.

The real-data partition uses random state 42. Selected real training seeds are 7012, 7024 and 7043, corresponding to run labels 11, 23 and 42. Plain ETM uses the same training seeds; only its last fit is historical. Tomotopy uses seeds 11, 23 and 42. Synthetic realization seeds are 11, 23 and 37, with training seeds 7012, 7024 and 7038 at both fitted topic counts. Within each real-data model, data partitions, vocabulary, embeddings and fitting settings are fixed across repetitions. No selected-model fit was rerun. The original plain-ETM fit ran under WSL2, whereas its two new repetitions ran on native Linux. Both use an RTX 5070 and matching recorded versions of the scientific packages, including PyTorch 2.10.0 with CUDA 12.8; the Python build and operating environment differ. The baseline bundle preserves both environment records. Its SD is therefore across fitted runs, not an isolated estimate of seed-only variation.

Generate the audited result fragments and run the scientific checks from the repository root:

```
python -S -m scripts.generate_contextual_reduction_review
python -m scripts.generate_motif_chemical_assessment_report
python -m scripts.generate_cross_model_overlap_report
pytest -q benchmarks/neural_ms2lda/tests
```

Compile both the main paper and this supplement from `docs/research/`, making their auxiliary files available through `TEXINPUTS`. Alternate `pdflatex` passes until cross-document references stabilize, then render and inspect every PDF page. The repository report README gives the full recipe. Probability, gradient and checkpoint-reload tests check that the reported equations match the chosen implementation. Per-run records remain in the research archive.

S8.3 Scope of the existing component evidence

In one paired real-data comparison of the closely related leave-one-out predecessor, removing channel balancing reduced useful motifs from 394 to 322. Restoring softmax in the within-model screen retained useful-motif breadth but produced much more diffuse mixtures, supporting entmax’s compactness role. Comparisons with context-free variants support the context pathway under the tested training settings, but do not establish that all spectra or all neural topic models require that pathway. Removing an entire evidence branch and removing only contextual mixing are different interventions and must not be conflated.

These are development comparisons, not factorial ablations of the final model. Whole-spectrum context preserved the discovery/compactness trade-off in three paired predecessor comparisons. The complete records are linked from the source inventory; they do not establish global model minimality.

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